

BUILDING FOOTPRINT EXTRACTION FROM REMOTE SENSING IMAGERY

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Abstract

Since the widespread availability of very high-resolution satellite imagery in the early 2000s, it has become possible to extract cartographic features from remotely sensed images. It is specifically important to extract building and other infrastructure footprints from satellite images of sufficient resolution, as these boundaries play an important role in the development of maps required by contexts like civil planning, emergency and disaster response, and monitoring illegal activity. In recent years the process of extracting building footprints has been taken over almost entirely by the application of neural network-based models to extract buildings from large datasets. Traditional spatial / spectral and LiDAR analysis methods paved the way for modern techniques. This paper reviews a variety of the most common ways buildings have been extracted from remotely sensed imagery in the past two decades. Each method is reviewed at a high level, advantages and disadvantages are discussed, and the most effective of the presented models are discussed in the conclusion.

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Introduction

Study Area

Remote sensing and GIS work together to integrate satellite imagery and spatial point, line, and polygon data for display on a map. In the context of urban planning, it is important to have well defined and accurate infrastructure footprints (e.g. buildings, roadways, dams, etc.). These features can be seen clearly in remote sensing imagery of very high resolution. Because of this, techniques exist to analyze remote sensing images of areas that contain infrastructure, and extract the boundaries of the infrastructure in a way where they can be displayed in GIS software as individual features. This is useful because generating building footprints by hand is extremely time consuming, and prone to user error. By approaching infrastructure footprint creation from a remote sensing standpoint, an analyst must only verify that the auto-generated footprints are in fact representative of what is on the ground. This process can also generate infrastructure footprints over large areas, which saves time and effort on the part of the analyst. Many parts of the world do not have good GIS data, so finding building footprints and information relating to the buildings themselves is a difficult task. By approaching this challenge with remote sensing and automated image analysis techniques via machine learning, spatial data can be generated for areas that have little GIS data available online.

Objectives

This paper defines the concept of building footprint extraction, discusses the most common approaches to building footprint extraction, and compares the most desirable remote sensing systems that can be used for this task, with respect to sensor design in terms of resolution and band coverage. The central question of this paper is “What remote sensing systems and post-processing techniques are most desirable for generating infrastructure footprints?”. This paper will provide an overview of the state-of-the-art techniques and workflows for footprint extraction from remote sensing imagery through the review of scholarly works and current projects in this study area.

Data

Required Data

One constant between traditional and modern footprint extraction methods is that they both employ satellite imagery as the basis for footprint extraction. Very High Resolution (VHR) images from various satellites and sensor platforms are usually used. Most of the images used are in the visible range, but multispectral and multiresolution images are employed by some models as the basis for analysis. Some datasets used by state-of-the-art models include the INRIA aerial imagery labeling dataset, IKONOS panchromatic imagery, LiDAR point clouds, and public satellite imagery datasets like Open Street Maps and Google Maps. Since the inception of using machine learning to extract building footprints from high resolution satellite imagery, an online competition called SpaceNet has challenged participants to submit their models for extracting building footprints. Participants receive access to the same datasets for different major cities around the world, and the data is typically retrieved from DigitalGlobe's WorldView-2 and WorldView-3 satellites (Pasquali et al., 2019). The SpaceNet competition serves as somewhat of a standard for researchers in this area to compete with and to test their own models against.

Methods

Traditional Methods

Historically the extraction of infrastructure footprints from remote sensing imagery is something that had to be done manually with supervised classification software like MultiSpec. Using this type of software, landcover types would be manually categorized, and “traditional” (pre-AI) classification models would be trained to classify the rest of the image by landcover type. The resulting classification would be brought into GIS software and turned into GIS features to represent landcover types that were determined by the models to be building footprints. Other traditional methods used to determine building footprints are edge detection and thresholding, which rely on a predefined set of parameters that tell the model how to discriminate building features based on things like color and texture. A LiDAR-based approach has also been used to determine 3-D building footprints from LiDAR-produced digital elevation models. Some modern techniques still employ LiDAR data for 3-D applications, and to isolate buildings roofs from parking lots because roofs have a nonzero Z-value.

Some problems with these traditional workflows is that they are very time-consuming, prone to user error, expensive, and typically focus on a very small study area. Image artifacts like shadows and overhanging trees make the problem even more challenging. These classification methods are not as accurate as modern techniques. These methods also struggle to identify all buildings in a study area, because environmental conditions such as lighting, buildings density, and vegetation cover make it difficult for models to determine the location of buildings precisely. Traditional (pre-AI) implementations of object detection in satellite imagery relies mostly on spatial / spectral image processing techniques to determine the edges of building features. Some traditional methods for extracting building footprints from high resolution satellite imagery are reviewed below.

Name: Automated 2-D Building Extraction from Multispectral Imagery

Description: This method uses multispectral imagery to identify buildings and their shadows. It uses “the differential morphological profile (DMP) of 1-m resolution panchromatic satellite imagery”. The DMP analysis is an image analysis technique that helps users to extract object information from imagery. The algorithm in this study is tested in IKONOS satellite imagery (Shackelford et al., 2004).

Name: Building Footprint Extraction and 3-D Reconstruction from LiDAR

Description: This study recognizes the need for an automated means of extracting building information from LiDAR data. A method for extracting building footprints and producing a 3-D reconstruction of buildings is presented. By extracting data points in LiDAR imagery that has a nonzero z-value, buildings are retrieved from LiDAR data. A filter is applied to the resulting extracted footprints to produce results that are sharper and have “better cartographic quality”. Building shapes were matched to the basic architecture types of buildings in the study area, and results were compared to a manually produced building reference dataset to test the accuracy of the algorithm (Haithcoat et al., 2001).

Modern Methods

Modern techniques have taken advantage of artificial intelligence and deep / machine learning models to produce superior results to traditional methods. The structure of these models is different from traditional classification models, and often employs some type of convolutional neural network to analyze imagery and determine the extent and location of building footprints. Images are classified by these models on a pixel-level, where each pixel is associated with a predefined class. Semantic segmentation is then carried out by the neural network model to classify the objects in the image. The use of these models in this application leads to more accurate results and saves analysts a lot of time. Large areas may be mapped out by feeding the relevant imagery to the model, improving the amount of area that can be mapped out when compared to traditional classification methods. Models may be trained with manually classified imagery that is known to be accurate, to improve the model's building-discrimination abilities. ESRI even offers a deep-learning model for use in ArcGIS Pro for the purposes of extracting building footprints from high-resolution satellite imagery. The ESRI model is routinely updated and may be used by analysts who have the required packages installed on their computer.

Because larger areas can be fed to the models for building extraction, a problem with these modern approaches is that manual error checking becomes a daunting task. A single user may be unable to review every single building that was extracted by the modern models. Another problem is that various neural network models developed for this application tend to ignore the smallest buildings because image downsampling done by the models loses resolution and can omit the smallest buildings from results. Some studies suggest hybrid approaches that analyze imagery from small scale to large scale to take advantage of the best parts of the most successful models (Guo et al., 2022). These works are reviewed in more detail in the next section.

In recent years, a deep-learning or hybrid model approach to building footprint extraction is vastly preferred to traditional methods. The most robust and successful models are based on convolutional neural network models designed specifically for the task of determining and extracting footprints for use in GIS software. The modern approaches are more accurate, can analyze much larger areas of land, and may be trained to improve over time. They are also much less expensive and save users a lot of time doing things manually. Modern implementations of object detection and deep learning neural networks for this application are diverse, and each has advantages and disadvantages. This section reviews a handful of the models used in recent years, by providing a brief overview for each. Many of these models are based in part on the famous U-Net convolutional neural network model, but some take a more novel approach.

Name: Course-to-Fine Boundary Refinement Network (CBR-Net)

Description: Complex building shapes, small buildings, and vegetation cover are major issues associated with some of the other neural network models. The results of these issues leads to inaccurate building footprints with blurry edges because semantic information about classes in the image cannot be preserved. The CBR-Net method uses multi-scale imagery to resolve some of the issues associated with the other methods by performing analysis at multiple image scales, refining the results of the extraction at each level in a course to fine manner. This allows small buildings not to be lost by model downsampling, effectively maps complicated (non-rectangular) building shapes, and produces sharp building edges (Guo et al., 2022).

Name: Multiple Attending Path Neural Network (MAP-Net)

Description: This model also uses a multi-scale approach to refining building footprints at different scales. Like the CBR-Net study, this study recognizes that other current models ignore small buildings, and the models that employ a multiscale approach still tend to produce discontinuous, patchy building footprints.

The MAP-Net model deals with these issues by learning spatial-localization preserved features that it identifies through the analysis. Because MAP-Net learns the location and shapes of features, it can recognize differences present in the same feature at different scales and preserve the resolution of the aspects of the feature determined to be an edge or building footprint. MAP-Net also deals with discontinuous and patchy building footprints by applying a ‘spatial pooling module’ to capture global dependencies of each feature and refine them to produce a single building where other models may produce a discontinuous building (Zhu et al, 2020).

Name: Semantic Segmentation (U-Net based model)

Description: This study uses data from the WorldView-3 Satellite Datasets to produce a semantic segmentation model from public GIS datasets like Open Street Maps and Google Maps. This study uses the SpaceNet building dataset provided at the IEEE Conference on Computer Vision and Pattern Recognition (2018). The study modifies the U-Net model to include “data augmentation, post-processing, and integration of GIS map data and satellite images” capabilities. The results of running this model on the mentioned data produces superior results to the other competitors in the SpaceNet Building Detection Competition, and to a purely U-Net based approach to building detection (Li et al., 2019).

Name: Multi-Resolution (U-Net based model)

Description: This study investigates the application of combining multiresolution imagery at early stages of analysis before passing the resulting imagery to a U-Net based model. Interestingly this study notes the challenge of separating buildings from parking lots or other building-shaped features from satellite imagery. By incorporating 3-D information about the features in satellite imagery using digital surface models, the U-Net based model can discriminate objects 'on the ground' from building rooftops. By combining imagery of different resolutions, building features are able to be maintained across different scales (Schuegraf et al., 2019).

Name: Spatial Residual Inception Network (SRI-Net)

Description: This study also incorporates a multiscale approach to building identification and extraction using a Spatial Residual Inception Network to "capture and aggregate multiscale contexts for semantic understanding by successively fusing multi-level features." This is a similar approach to the MAP-Net model. An advantage to this model is that it can detect large buildings that might be omitted or discontinuous when identified by other non-multiscale models (Penghua et al., 2019).

Name: Deep Supervision Convolutional Network (DS-Net)

Description: The DS-Net also incorporates a multiscale approach, but the methods for model training are different. This approach uses "deep supervision with an extra lightweight encoder" which allows the network to learn what different building features look like at different scales. This is effectively a novel way to train the model to recognize buildings, and like the SRI-Net model, a multiscale approach is used to "aggregate features and generate final building predictions" (Guo et al, 2021).

Name: Update Building Footprints with saliency-guided edge-preservation (SG-EPUNet)

Description: This approach focuses on updating existing infrastructure databases in a semi-supervised manner, using a neural network to identify new buildings over time. This neural network-based model is intended to update infrastructure databases with little manual intervention. SG-EPUNet required “manual annotations for only 20% of the buildings”. The results of the study that introduces this model suggest that it is robust and scalable, because it maintains accuracy with few annotations over large study areas. This study is important because it presents a method for automating the update of infrastructure databases with accurate information and little manual intervention, to keep track of infrastructure in rapidly developing urban areas (Guo et al., 2021).

Conclusions

The results of modern neural-network-based approaches to building footprint extraction are superior to traditional methods. The amount of imagery that can be processed and the accuracy of the results leads researchers in this area to prefer neural network-based approaches to building footprint extraction. Within the modern methods, the SRI-Net approach appears to be the most robust and scalable. While other models perform similarly, there are some drawbacks associated with them such as the loss of smaller buildings, and the inability to incorporate multiresolution imagery or 3-D data into the models.

The rapid urbanization of cities requires a method for keeping databases up to date quickly and efficiently. Future research in this area should begin to take an approach like the SG-EPUNet to update building and infrastructure databases with little manual intervention. It would be interesting to see the development of neural network-based models which, like the MAP-Net and CBR-Net, focus on merging the best aspects of other state-of-the-art models to minimize drawbacks associated with the modern methods. Future work should also focus on the interoperability of models and the datasets they use. This would allow end-users to have more flexibility in terms of their building footprint extraction approach, and in turn may lead to more accurate results.

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